



Anuj Vishwakarma

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech	Indian Institute of Technology, Guwahati	7.79 (Current)	2022-Present
Senior Secondary	CBSE Board	92.8%	2022
Secondary	CBSE Board	93.6%	2020

EXPERIENCE

- Tollring India Pvt. Ltd.** May 2025 - July 2025
Software Intern Noida, India
 - Developed a multi-reasoning agentic AI framework using **CrewAI**, **NL2SQL**, and **MCP server** for cross-file query answering, attained **65% accuracy** on the "Hard" query set of the DABStep benchmark.
 - Integrated MCP with **ElasticSearch** to speed up call metadata retrieval and improve **RAG model** accuracy.
 - Prototyped modular agents for **4Com** to auto-evaluate **50+ sub-KPIs**, enhancing call review efficiency.
- Indian Institute of Technology Guwahati** April 2024 - June 2024
Research Intern under Dr. Hrishikesh Sharma, Associate Professor, IITG Guwahati, Assam, India
 - Built a **DNN** to predict concrete strength ($R^2 = 0.88$), utilized **Keras Tuner** for hyperparameter optimization.
 - Generated and optimized new concrete configurations to achieve target strength with minimal **Global Warming Potential (GWP)**, proposed theoretical mix designs capable of reducing GWP by up to **50%**.

PROJECTS

- SeeForMe - Image to Speech Captioning System** May 2025 - June 2025
Personal Project GitHub
 - Built a **CNN-GRU** model with **attention** for image captioning to aid visually impaired users.
 - Executed comprehensive **preprocessing** and **visualization** of multimodal data, leveraged **InceptionV3** for feature extraction, and enhanced caption generation using **Greedy** and **Beam Search** decoding strategies.
 - Converted generated captions into natural audio using **TTS API**, making visual content accessible through sound.
- GenreFy - Music Genre Classification** May 2025 - June 2025
Personal Project GitHub
 - Predicted genres from GTZAN dataset using handcrafted features, performed **EDA** on audio files, and implemented **KNN**, **SVM**, **Logistic Regression**, **Gaussian NB** and **Soft-Voting Ensemble** with **Cross-Validation**.
 - KNN achieved **88%** accuracy, highlighting the effectiveness of classical ML over complex deep learning architectures.
- News Sentiment Analysis** May 2024 - June 2024
Coding Club, IIT Guwahati, Individual Project GitHub
 - Developed a machine learning model for news sentiment classification using **NLP preprocessing** and **Vader** for data labeling, utilized **embedding** and **dense layers**, and deployed via **Django** for an end-to-end workflow.
 - Achieved **86.83%** accuracy with macro average **precision**, **recall**, and **F1-scores** around **0.865**.
- AI Image Classifier** May 2024 - June 2024
Coding Club, IIT Guwahati, Team Project GitHub
 - Built a **VGG16-inspired CNN** to classify Real vs. AI-generated images with **convolutional**, **pooling**, and **dropout layers**, achieved an overall accuracy of **94%**, with macro average **precision**, **recall**, and **F1-scores** all at **0.94**.

TECHNICAL SKILLS

- Programming:** C/C++, Python
- Tools/Frameworks:** Scikit-Learn, Keras, TensorFlow
- Database Management:** MySQL
- Data Analysis:** SQL, Numpy, Pandas, Matplotlib

KEY COURSES TAKEN

- Computer Science & Mathematics:** Data Structures and Algorithms, Computing Laboratory, Introduction to Deep Learning Internals, Linear Algebra, Calculus, Real Analysis, Complex Analysis.
- MOOCs:** Machine Learning, Structuring ML Projects, Improving Deep Neural Networks, Advanced Learning Algorithms, Convolutional Neural Networks, Unsupervised Learning.

ACHIEVEMENTS & EXTRACURRICULARS

- Leetcode**, Secured a Global **Rank 1853** among 35000+ participants in Biweekly contest 165 (Handle : anuj120377).
- JEE Advanced 2022**, Secured an All India open category rank of **9528** among 0.2 million candidates.
- JEE Mains 2022**, Scored **98.83 percentile** among 2 million candidates.
- IISER 2022**, Secured an All India rank of **1060** in IAT 2022.
- NDA**, Cleared the NDA written exam three times and attended the **SSB interview** at Dehradun in 2022.
- Cricket:** Participated in **Inter IIT Sports**, **Gold** at Spirit (IITG) & Udghosh (IITK), Champions of 4th Nilay Dutta **Inter-Varsity Cricket** Tournament organized by **Assam Cricket Board**.
- Hockey**, Won **Gold Medal** in Spardha 2024.
- Music**, Studied Indian Classical Music from **Akhil Bhartiya Gandharva Mahavidyalaya Mandal**.